

NAME: _____

PERIOD: _____

DATE: _____

Same Responses, Different Questions

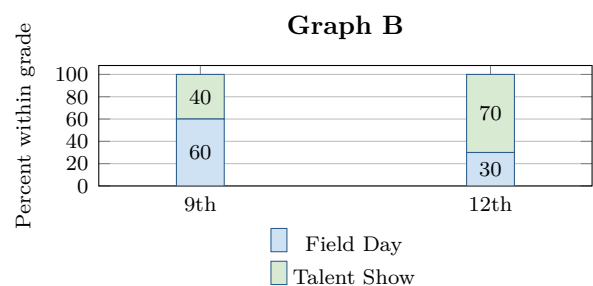
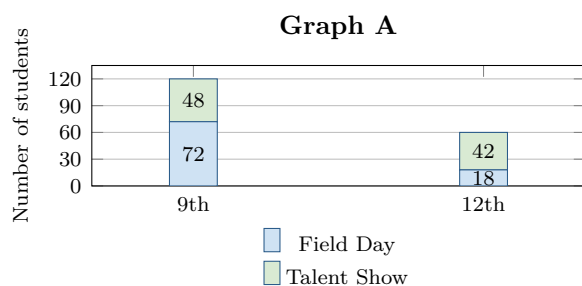
AP Statistics · Unit 2 · Topic 2.1 · B: 2026-10-15 · E: 2026-10-28

[STAR] TODAY'S PROBLEM

Suppose a student council asks 120 ninth graders and 60 twelfth graders whether they prefer Field Day or a Talent Show for the spring celebration. The same responses are shown in two graphs.

Andre: *Graph B hides how many people there are because both bars reach 100%.*

Leila: *Graph A hides the relationship because the grade totals are different.*



FIRST TAKE — before the video (5 min)

For the question *Is celebration preference related to grade level?*, which graph would you use first? Name one visible feature.

[BOOK] MATCH THE DISPLAY TO THE QUESTION (keep on the page)

A **count bar graph** preserves how many observations are in each category and group. A **segmented bar graph** stacks each group's conditional relative frequencies to 100%. Equal total heights remove the effect of unequal group sizes, so corresponding segments compare proportions. Different conditional distributions indicate an association. A segmented graph needs group totals before it can recover counts; a count graph needs division before it directly shows within-group percentages.

Finish after the video:

DOK 1 (a) From Graph A, state each grade's total surveyed and the count in each preference category.

DOK 2 (b) Compute the conditional distribution of preference within each grade. Describe the largest contrast between the grades.

DOK 3 (c) For deciding whether preference is related to grade level, choose the more useful graph and explain how equal-height bars settle the comparison. State what Graph A is useful for. Then give one question that Graph A cannot answer *directly from bar lengths alone* and one question that Graph B cannot answer from the display.

Frame: Graph _____ better answers the relationship question because equal-height bars let me compare _____.

Frame: Graph A preserves _____, while Graph B cannot tell me _____.

EXIT — turn this in

One thing the video changed about my first take: _____